Heavy Suppliers Warehouse Dataset — Data Profiling & Structure Analysis

Corrected 2026-10-26: duplicate-ID counts read as 196 invoice IDs and 201 payment IDs; P002 margin note added

Profiling of 12 CSV datasets covering branches, customers, suppliers, products, inventory, purchase orders, sales orders, invoices, payments and stock movements.

Metric	Value
Datasets analyzed	12
Total rows across all files	605,141
Total columns across all files	134
Exact duplicate rows found	0
Foreign-key relationships tested	17
Orphaned foreign-key rows found	0

The archive contains 12 CSV files representing a heavy-equipment spare-parts distribution business across six branches. The tables form a typical operational schema: dimension tables (branches, customers, suppliers, products) and transactional/fact tables (purchase orders, sales orders, invoices, payments, stock ledger, inventory master). Row and column counts below were computed directly from the complete CSV files (not a sample). “Missing cells” counts every null value across every column in the file.

1. Dataset Overview

Dataset	Rows	Columns	Missing Cells	Exact Duplicate Rows
branches.csv	6	13	0	0
customers.csv	500	15	0	0
inventory_master.csv	180	8	0	0
invoices.csv	18,033	10	0	0
payments.csv	19,257	5	0	0
products.csv	30	23	0	0
purchase_orders_header.csv	24,000	10	2370	0
purchase_orders_lines.csv	155,495	9	0	0
sales_orders_header.csv	20,000	11	0	0
sales_orders_lines.csv	130,402	9	0	0
stock_ledger.csv	237,230	9	0	0
suppliers.csv	8	12	0	0

Entity relationships (keys observed):

- branches (branch_id) ← customers, invoices, sales_orders_header, purchase_orders_header, inventory_master, stock_ledger
- customers (customer_id) ← sales_orders_header, invoices
- suppliers (supplier_id) ← purchase_orders_header
- products (product_id) ← inventory_master, sales_orders_lines, purchase_orders_lines, stock_ledger
- sales_orders_header (so_id) ← sales_orders_lines, invoices
- purchase_orders_header (po_id) ← purchase_orders_lines
- invoices (invoice_id) ← payments

Note on branches.csv: confirmed exactly 6 rows × 13 columns, with zero missing cells and zero exact duplicate rows.

The only “missing cells” in the whole archive are the 2370 nulls in purchase_orders_header.received_date — every other file is 100% complete.

2. Column-Level Data Profiling

For every table, each column: data type, non-null count, missing count, and the number of distinct values (unique) were computed. High uniqueness relative to row count indicates an identifier-like column.

2.1 branches.csv

6 rows × 13 columns

Column	Data Type	Non-null	Missing	Unique
branch_id	str	6	0	6
branch_name	str	6	0	6
city	str	6	0	6
state	str	6	0	6
region	str	6	0	4
warehouse_type	str	6	0	4
warehouse_capacity	str	6	0	6
service_center_available	str	6	0	2
manager_id	int64	6	0	6
total_employees	int64	6	0	6
avg_monthly_revenue	int64	6	0	6
monthly_operational_cost	int64	6	0	6
market_demand_index	int64	6	0	4

2.2 customers.csv

500 rows × 15 columns

Column	Data Type	Non-null	Missing	Unique
customer_id	str	500	0	500
customer_type	str	500	0	5
industry_segment	str	500	0	6
city	str	500	0	21
state	str	500	0	16
pincode	int64	500	0	500
region	str	500	0	6
branch_id	str	500	0	6
credit_limit	int64	500	0	500
current_balance	int64	500	0	499
payment_terms	str	500	0	5
customer_since	str	500	0	467
last_purchase_date	str	500	0	449
total_purchase_value	int64	500	0	500
customer_rating	int64	500	0	5

2.3 inventory_master.csv

180 rows × 8 columns

Column	Data Type	Non-null	Missing	Unique
product_id	str	180	0	30
branch_id	str	180	0	6
opening_stock	int64	180	0	127

Column	Data Type	Non-null	Missing	Unique
reorder_level	int64	180	0	75
safety_stock	int64	180	0	63
max_stock	int64	180	0	148
current_stock	int64	180	0	180
warehouse_bin	str	180	0	95

2.4 invoices.csv

18,033 rows × 10 columns

Column	Data Type	Non-null	Missing	Unique
invoice_id	str	18,033	0	17,836
so_id	str	18,033	0	18,033
customer_id	str	18,033	0	500
branch_id	str	18,033	0	6
invoice_date	str	18,033	0	2,203
due_date	str	18,033	0	2,235
total_order_value	int64	18,033	0	16,248
total_gst_amount	float64	18,033	0	16,951
grand_total	float64	18,033	0	16,938
payment_status	str	18,033	0	3

2.5 payments.csv

19,257 rows × 5 columns

Column	Data Type	Non-null	Missing	Unique
payment_id	str	19,257	0	19,055
invoice_id	str	19,257	0	16,036
payment_date	str	19,257	0	2,243
payment_amount	float64	19,257	0	18,778
payment_method	str	19,257	0	5

2.6 products.csv

30 rows × 23 columns

Column	Data Type	Non-null	Missing	Unique
product_id	str	30	0	30
product_name	str	30	0	30
category	str	30	0	15
machine_type	str	30	0	6
brand	str	30	0	6
model_compatibility	str	30	0	18
unit_cost	int64	30	0	30
unit_price	int64	30	0	28
margin_percentage	float64	30	0	30
gst_rate	int64	30	0	2
weight_kg	float64	30	0	29
dimensions_cm	str	30	0	29
material_type	str	30	0	10

Column	Data Type	Non-null	Missing	Unique
warranty_months	int64	30	0	5
reorder_level	int64	30	0	20
safety_stock	int64	30	0	17
max_stock_level	int64	30	0	19
lead_time_days	int64	30	0	19
criticality_level	str	30	0	3
usage_frequency	str	30	0	3
uom	str	30	0	3
last_purchase_price	int64	30	0	30
last_purchase_date	str	30	0	30

2.7 purchase_orders_header.csv

24,000 rows × 10 columns

Column	Data Type	Non-null	Missing	Unique
po_id	str	24,000	0	24,000
supplier_id	str	24,000	0	8
branch_id	str	24,000	0	6
order_date	str	24,000	0	2,192
expected_delivery_date	str	24,000	0	2,207
received_date	str	21,630	2370	2,207
po_status	str	24,000	0	2
total_cost	float64	24,000	0	24,000
total_gst_amount	float64	24,000	0	24,000
grand_total	float64	24,000	0	24,000

2.8 purchase_orders_lines.csv

155,495 rows × 9 columns

Column	Data Type	Non-null	Missing	Unique
po_id	str	155,495	0	24,000
line_number	int64	155,495	0	12
product_id	str	155,495	0	30
quantity	int64	155,495	0	281
unit_cost	float64	155,495	0	142,292
gst_rate	int64	155,495	0	2
line_total	float64	155,495	0	155,224
gst_amount	float64	155,495	0	155,254
line_grand_total	float64	155,495	0	155,269

2.9 sales_orders_header.csv

20,000 rows × 11 columns

Column	Data Type	Non-null	Missing	Unique
so_id	str	20,000	0	20,000
customer_id	str	20,000	0	500
branch_id	str	20,000	0	6
order_date	str	20,000	0	2,192

Column	Data Type	Non-null	Missing	Unique
delivery_date	str	20,000	0	2,204
order_status	str	20,000	0	2
payment_terms	str	20,000	0	5
total_order_value	int64	20,000	0	17,905
total_gst_amount	float64	20,000	0	18,749
grand_total	float64	20,000	0	18,735
sales_channel	str	20,000	0	4

2.10 sales_orders_lines.csv

130,402 rows × 9 columns

Column	Data Type	Non-null	Missing	Unique
so_id	str	130,402	0	20,000
line_number	int64	130,402	0	12
product_id	str	130,402	0	30
quantity	int64	130,402	0	20
unit_price	int64	130,402	0	28
gst_rate	int64	130,402	0	2
line_total	int64	130,402	0	534
gst_amount	float64	130,402	0	534
line_grand_total	float64	130,402	0	534

2.11 stock_ledger.csv

237,230 rows × 9 columns

Column	Data Type	Non-null	Missing	Unique
movement_id	str	237,230	0	237,230
product_id	str	237,230	0	30
branch_id	str	237,230	0	6
movement_type	str	237,230	0	3
movement_date	str	237,230	0	2,218
quantity	int64	237,230	0	300
reference_type	str	237,230	0	3
reference_id	str	237,230	0	43,709
running_balance	int64	237,230	0	98,171

2.12 suppliers.csv

8 rows × 12 columns

Column	Data Type	Non-null	Missing	Unique
supplier_id	str	8	0	8
supplier_name	str	8	0	8
supplier_type	str	8	0	3
product_category	str	8	0	3
city	str	8	0	8
province	str	8	0	6
region	str	8	0	3
pincode	int64	8	0	8

Column	Data Type	Non-null	Missing	Unique
lead_time_days	int64	8	0	7
reliability_score	int64	8	0	3
import_duty_rate	int64	8	0	4
china_tax_id	str	8	0	8

3. Data Quality & Consistency Findings

Every check below was re-run directly against the raw CSVs so the counts here are reproducible.

Check	Dataset / Field	Count	Interpretation
Missing values	purchase_orders_header.received_date	2,370	All null received_date rows correspond exactly to the 2,370 cancelled POs — expected missingness, not an error.
Duplicate business key	invoices.invoice_id	196 IDs reused across 393 rows	18,033 rows carry only 17,836 distinct IDs (197 surplus rows). Do not use invoice_id alone as a unique key.
Duplicate business key	payments.payment_id	201 IDs reused across 403 rows	19,257 rows carry only 19,055 distinct IDs (202 surplus rows). Do not use payment_id alone as a unique key.
Referential integrity	17 tested foreign-key relationships	0 orphans	Every child-to-parent key relationship tested is fully intact.
Date chronology	PO received_date vs expected_delivery_date	971 early / 3,267 late	Of 21,630 received POs: 971 arrived before the expected date, 17,392 arrived exactly on it, and 3,267 arrived after it. None arrived before their own order_date.
Range/business rule	inventory_master.current_stock vs max_stock	180/180 rows	current_stock exceeds max_stock in every row, running 331–1369x opening_stock. Needs validation against stock_ledger before use.
Derived-field consistency	products.margin_percentage	29/30 match one formula	margin_percentage matches a cost-basis markup formula, (price−cost)/cost×100, for 29 of 30 products. At profiling time P002 was stored as 37.3 and fit neither formula (see correction note below).
Calculation consistency	PO & SO line/header totals	0 mismatches	line_total, gst_amount, line_grand_total and header grand_total all reconcile exactly.
Payment reconciliation	invoices.payment_status vs paid amount	84 rows	Comparing each invoice’s aggregated payments to its grand_total, 84 invoices have a payment_status that doesn’t match the amount actually paid.

Check	Dataset / Field	Count	Interpretation
Payment reconciliation	aggregated payments > grand_total	306–510 rows	306 invoices are overpaid by more than 1%; 510 show any overage at all. The exact count is sensitive to how the 393 duplicate invoice_id rows are resolved before aggregating — resolve that first.

[CRITICAL] inventory_master.current_stock is implausibly large. current_stock ranges from 88,853 to 121,013 units per branch/product, while opening_stock, reorder_level, safety_stock and max_stock are all in the hundreds (e.g. max_stock tops out around 300–500). The ratio of current_stock to opening_stock ranges from roughly 330x to 1,369x across all 180 rows, and current_stock exceeds max_stock in 100% of rows (180/180). This looks like a unit or generation error in this column rather than a real stock level, and any dashboard using it (stock-on-hand, days-of-cover, reorder alerts) will be wrong until it is corrected.

[CRITICAL] Non-unique primary keys in invoices and payments. invoice_id has 196 reused IDs (393 rows share an ID with another row) and payment_id has 201 reused IDs (403 rows affected). The earlier figures of 197 and 202 are the surplus rows (393 − 196 and 403 − 201), not the number of reused IDs. These are not exact duplicate rows — they are different transactions that were assigned the same ID, so invoice_id and payment_id cannot be relied on as unique keys in joins or lookups without first deduplicating or re-keying.

[MODERATE] 306 invoices show cumulative overpayment. Summing all payments per invoice and comparing to grand_total shows 306 invoices where total payments received exceed the invoice's grand_total by more than 1%. This may reflect legitimate advance payments/credit notes, but is worth reconciling against payment_status.

[INFO] purchase_orders_header.received_date is null for all cancelled POs. 9.88% of purchase_orders_header rows (2,370) have a null received_date. This lines up exactly with the count of po_status = 'Cancelled' orders — a consistent and expected pattern, not a data error.

[INFO] Header/line financial totals reconcile perfectly. grand_total = total_order_value/total_cost + total_gst_amount holds (within rounding) for all rows in invoices, sales_orders_header and purchase_orders_header — no mismatches found.

[INFO] Date logic is internally consistent. No rows were found where a due_date precedes its invoice_date, a delivery_date precedes its order_date, or a received_date precedes its order_date.

[INFO] Free-text/mixed-format fields. branches.warehouse_capacity stores values as text with a unit suffix (e.g. '45230 sqft') rather than a plain number, and products.dimensions_cm packs three measurements into one string (e.g. '40x25x25'). Both are usable for display but need parsing before numeric analysis.

Correction notes (2026-10-26).

- **P002 margin:** margin_percentage is a markup on cost, (price − cost) / cost × 100. P002 (cost 450, price 720) gives 60.0, so it follows the same formula as the other 29 products; the 37.3 value seen at profiling time was the outlier and has since been corrected to 60.0.
- **inventory_master.current_stock vs the ledger:** the latest ledger balance matches current_stock on 166 of 180 positions; 14 differ by 16–515 units.

4. Structural Relationships checks

Every foreign-key style relationship implied by the schema was tested by comparing child-table key values against the parent table's key values. Result: all 17 relationships tested are fully intact — zero orphaned rows across the entire dataset. This is a strong sign of consistent key generation across files.

Parent Table	Parent Key	Child Table	Child Key
branches.csv	branch_id	customers.csv	branch_id
products.csv	product_id	inventory_master.csv	product_id
branches.csv	branch_id	inventory_master.csv	branch_id
sales_orders_header.csv	so_id	invoices.csv	so_id
customers.csv	customer_id	invoices.csv	customer_id
branches.csv	branch_id	invoices.csv	branch_id

Parent Table	Parent Key	Child Table	Child Key
invoices.csv	invoice_id	payments.csv	invoice_id
suppliers.csv	supplier_id	purchase_orders_header.csv	supplier_id
branches.csv	branch_id	purchase_orders_header.csv	branch_id
purchase_orders_header.csv	po_id	purchase_orders_lines.csv	po_id
products.csv	product_id	purchase_orders_lines.csv	product_id
customers.csv	customer_id	sales_orders_header.csv	customer_id
branches.csv	branch_id	sales_orders_header.csv	branch_id
sales_orders_header.csv	so_id	sales_orders_lines.csv	so_id
products.csv	product_id	sales_orders_lines.csv	product_id
products.csv	product_id	stock_ledger.csv	product_id
branches.csv	branch_id	stock_ledger.csv	branch_id

5. Key Categorical Distributions

Invoice payment_status (18,033 invoices)

Value	Count	Share
Paid	12,536	69.5%
Partially Paid	3,660	20.3%
Unpaid	1,837	10.2%

Sales order order_status (20,000 orders)

Value	Count	Share
Delivered	18,033	90.2%
Cancelled	1,967	9.8%

Sales order sales_channel

Value	Count	Share
Field Sales	5,036	25.2%
Dealer Network	5,028	25.1%
Counter Sale	5,014	25.1%
Online	4,922	24.6%

Purchase order po_status (24,000 orders)

Value	Count	Share
Received	21,630	90.1%
Cancelled	2,370	9.9%

Customer customer_type (500 customers)

Value	Count	Share
Government	114	22.8%
Corporate	104	20.8%
Dealer	99	19.8%
Fleet Owner	95	19.0%
Retail	88	17.6%

Customer industry_segment

Value	Count	Share
Mining	98	19.6%
Logistics	96	19.2%
Construction	90	18.0%
Infrastructure	79	15.8%
Manufacturing	70	14.0%
Rental	67	13.4%

Stock ledger movement_type (237,230 movements)

Value	Count	Share
IN	127,611	53.8%
OUT	107,165	45.2%
ADJUSTMENT	2,454	1.0%

6. Profiling Conclusion & Recommendations

The warehouse data is structurally sound: every table has a clear master, header, line, inventory, invoice, payment or ledger role, and all 17 tested foreign-key relationships are fully intact with zero orphan references. Financial arithmetic (line totals, GST, and header/line reconciliation) is exact throughout. The issues below should be handled in a data-cleaning stage rather than silently deleted or overwritten:

- Resolve the 393 rows in invoices.csv (196 reused IDs) and 403 rows in payments.csv (201 reused IDs) that share a repeated invoice_id / payment_id before using either as a unique key.
- Validate inventory_master.current_stock: it exceeds max_stock in all 180 rows and runs 331–1369x opening_stock — cross-check it against stock_ledger.running_balance for the same product/branch before treating it as reliable.
- Review the 971 purchase orders received earlier than their expected_delivery_date, and the 3,267 received later, to confirm these reflect genuine early/late deliveries rather than date-entry errors.
- Confirm the intended definition of products.margin_percentage: 29 of 30 products matched a cost-basis markup formula, (price−cost)/cost×100, at profiling time. The definition is now confirmed as markup on cost, and P002 has been corrected to 60.0.
- Decide how to resolve duplicate invoice_id values before reconciling payments — the overpayment count shifts between 306 and 510 rows depending on that choice, and 84 invoices' stored payment_status don't match their paid amount either way.
- Treat the 17 foreign-key relationships and the header/line financial reconciliations as validated — they can be relied on as-is for joins and financial rollups.